BTS–VII(S)–05–24–2394 Reg. No. | | | | | | | | |

B

B.Tech. Degree VII Semester Supplementary Examination May 2024

ME 19-205-0706 AUTOMOBILE ENGINEERING

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome
On successful completion of the course, the students will be able to:

CO1: Compare the working of various types of automobiles.
CO2: Learn the functions and working of different engine components.
CO3: Appreciate the use of alternate fuels.
CO4: Understand the latest technology used in the field of automobile.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	What do you mean by autothermic piston? Explain.	3	L2	2	6,7
	(b)	What is meant by wheel balancing? Explain.	3	L2	4	6,8
	(c)	What are the different types of lubrication systems available for automobiles? Explain.	3	L2	2	8,9
	(d)	What are the advantages of using bio-diesel as an alternate fuel?	3	L2	3	6
	(e)	Explain the basic principle behind the working of an epicyclic gearbox.	3	L2	2	6,7
	(f)	Explain the working of an overdrive with the help of a neat sketch.	3	L2	2	6,7,8
	(g)	Which type of batteries are used in electric vehicles? Also state it's limitations.	3	L2	4	6,7
	(h)	Explain the flywheel based energy storage device.	3	L2	3	6,8

PART B

		$(4 \times 12 = 48)$				
II.	(a)	Explain the constructional terminology of crank shaft with a neat sketch. What are the manufacturing methods and materials used for crank shaft?	8	L2	2	8,9
	(b)	What are the types of bearings and it's materials used for engine construction? Explain.	4	L2	2	8,9
		OR				
III.		What do you mean by valve timing? Explain the valve timing diagram of an SI engine and also discuss the advantages and disadvantages of valve lead, valve lag and valve overlap.	12	L2	4	7,8,9
IV.		What are the defects in a simple carburetor for different working conditions? Explain the working of an automatic choke with a neat sketch.	12	L2	2	6,8, 10
		OR				
V.	(a)	What are the different types of air cooling methods available for an IC engine? Mention it's advantages and disadvantages.	2	L3	2	9,10
	(b)	Explain the working of a thermosyphon system and pump circulation cooling system with neat sketches.	10	L2	2	8,9

(P.T.O.)

			Marks	BL	CO	PO
VI.		Explain the working of torque converter and discuss how freewheel unit is working with torque converter to reduce high speed instability.	12	L2	1	8,9, 10
		OR				
VII.	(a)	What are the factors considered for wheel alignment? Explain.	5	L2	4	8,9, 11
	(b)	Explain the working of a constant mesh gear box with neat sketches. Also discuss it's advantages and disadvantages.	7	L2	2	8,9
VIII.		Explain the different types of hybrid systems used in vehicles with neat sketches. Also mention its advantages and disadvantages.	12	L2	4	9,10
		OR				
IX.		Discuss the working of an electric car with the help of a block diagram. Also compare it with a fuel cell vehicle.	12	L2	4	9,10

Blooms's Taxonomy Levels
L2 – 90%, L3 – 10%.

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